Cyber Security

Subject: Computer Science | NoteShare Library

CYBER SECURITY

1. CIA TRIAD
Confidentiality
Integrity
Availability

2. CRYPTOGRAPHY
Symmetric: AES, DES
Asymmetric: RSA, ECC
Hash: SHA-256, MD5
Digital signatures
PKI, Certificates

3. ATTACKS
Phishing
SQL Injection
XSS (Cross-site scripting)
CSRF
MITM
DDoS
Ransomware
Social engineering
Password attacks

4. DEFENSE
Firewalls
IDS/IPS
VPN
Antivirus
MFA
Encryption
Regular updates
Least privilege

5. AUTHENTICATION
Passwords (bcrypt, argon2)
OAuth 2.0
JWT
Biometrics
SSO

6. NETWORK SECURITY
HTTPS (TLS)
IPsec
WPA3
Port security
Network segmentation

7. WEB SECURITY
OWASP Top 10
Input validation
Output encoding
CSRF tokens
CSP headers
SameSite cookies